Machine Learning-Based Heuristic Functions in Large-Scale Urban Routing Problems: Bidirectional and Hierarchical A* Route Optimization in the Case of Istanbul

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Abstract

Optimal route planning in metropolises with large-scale, asymmetrical topologies and bottlenecks (e.g., bridges, tunnels) incurs high computational costs for classical graph search algorithms. This study presents a neural network-assisted, hierarchical, and bidirectional A* route optimization system designed for Istanbul's complex road network. In the developed architecture, OpenStreetMap data is divided into two hierarchical layers, "Express" and "Arterial," to narrow the search space. Instead of the classical Haversine heuristic, a Multi-Layer Perceptron (MLP) model trained with ground-truth A* costs is integrated. To maximize inference speed, the model is exported in ONNX format with INT8 quantization. For multi-stop routes, an asymmetric cost matrix is generated using the neural network, and the Traveling Salesperson Problem (TSP) is solved by hybridizing Nearest Neighbor and 2-opt algorithms. Test results demonstrate that bidirectional search and hierarchical layering significantly reduce the search space, while the neural network-based heuristic achieves the optimal result faster by expanding substantially fewer nodes compared to classical methods.

Keywords: A* Algorithm, Machine Learning, ONNX, Traveling Salesperson Problem, Route Optimization, Istanbul.

1. Introduction

Developing smart city systems and modern logistics networks requires high-speed, dynamic route planning solutions for urban transportation [1]. In cities like Istanbul, which spans two continents and where intercontinental transitions are provided via a limited number of arteries (suspension bridges, the Eurasia Tunnel, etc.), classical algorithms such as Dijkstra [2] or standard A* [3] struggle to offer real-time performance due to an expanding search space. In a standard search tree with depth d and branching factor b, the time complexity reaches exponential levels of $\mathcal{O}(b^d)$.

This paper combines three fundamental approaches to narrow the search space and minimize execution time: hierarchical graph layering, bidirectional search, and a machine learning (ML)-based heuristic function [4]. The proposed "Istanbul Route Optimizer" system not only finds the shortest path between a source and destination but also incorporates a complex TSP solver to determine the optimal sequence for visiting N intermediate waypoints.

2. Literature Review

Route optimization and pathfinding algorithms are among the most established research areas in computer science. The A* algorithm, introduced by Hart, Nilsson, and Bertram [3], has been the industry standard for many years due to its guarantee of optimality. Bidirectional search techniques, later developed by Pohl [6], theoretically reduced the search space to $\mathcal{O}(b^{d/2})$ by proceeding simultaneously from the start and target nodes.

In recent years, with the rise of deep learning and Graph Neural Networks (GNN), classical heuristic algorithms have begun to be replaced by data-driven models. Pointer Networks proposed by Vinyals et al. [11] and the attention-based routing models by Kool et al. [12] proved the power of machine learning in combinatorial optimization problems [13].

Particularly in post-2022 literature, the integration of reinforcement learning and deep learning in Vehicle Routing Problems (VRP) has accelerated [15, 16, 18]. In 2024 and 2025, AI-assisted bidirectional search algorithms for large-scale and Multi-Agent Path Finding (MAPF) systems dominated the literature. While Li et al. (2024) [20] and Wang et al. (2025) [24] integrated lightweight versions of the bidirectional A* algorithm into autonomous vehicles, Zhang et al. (2025) [23] and Skrynnik et al. (2026) [19] optimized search nodes using temporal-spatial information aggregation techniques with Graph Attention Networks (GAT). Furthermore, machine learning-assisted green vehicle routing solutions in smart logistics networks are rapidly becoming widespread [21, 22]. This study contributes to the existing body of knowledge by adapting the most current ML heuristic approaches [25] to a specific and highly complex metropolitan topology.

3. Methodology and System Architecture

The proposed system is built on a multi-modular data pipeline extending from raw map data processing to user interface rendering.

3.1. Network Topology and Data Processing

The main road network of Istanbul was extracted from OpenStreetMap (OSM) using the Overpass API and the OSMnx library [5]. The dataset was filtered to include only main arteries (motorway, trunk, primary, secondary), explicitly omitting low-priority residential roads. To prevent routing errors caused by dead ends and disconnected one-way segments, the graph was cleaned using a Strongly Connected Component (SCC) analysis.

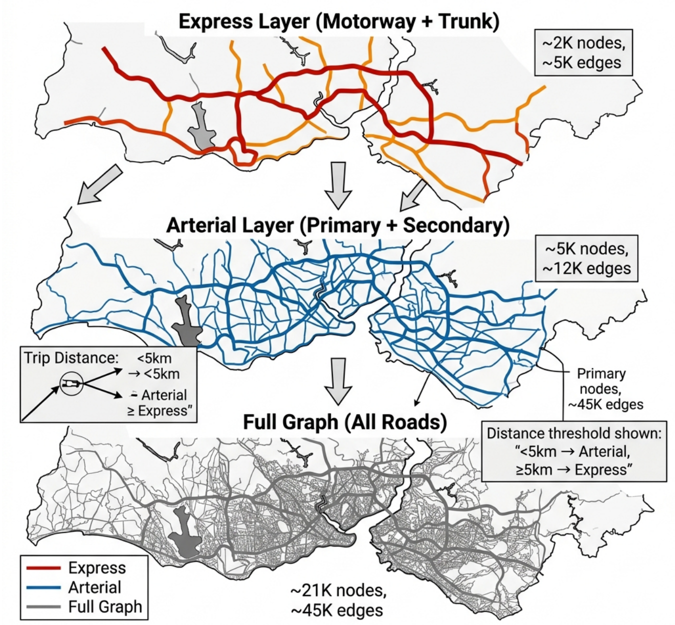
To enhance system performance, the network is divided into two hierarchical layers:

Express Layer: Contains only motorways and trunk roads. Used primarily for searches with a straight-line distance greater than 5 kilometers.

Arterial Layer: Includes primary and secondary roads. Preferred for detailed, intra-city searches shorter than 5 kilometers.

Layer	Road Types	Number of Nodes	Number of Edges	Usage Distance	Speed Advantage
Express	motorway, trunk	~2.000	~5.000	≥ 5 km	~10x
Arterial	primary, secondary	~5.000	~12.000	< 5 km	~4x
Full (Fallback)	All Way	~21.000	~45.000	Fallback	1x (referans)

Figure 2: Hierarchical Graph Layer Structure



3.2. Bidirectional Hierarchical A* Algorithm

Feature	Dijkstra	One-Way A*	Bidirectional A*	A* + Hierarchy
Time Complexity	O(V log V + E)	O(b^d)	O(b^(d/2))	O(b^(d/4)) (layer)
Optimality	✅ Guarantee	✅ Guarantee (admissible h)	✅ Guarantee (Pohl, 1971)	✅ Guarantee
Intuitive Function	❌ No	✅ Haversine	✅ Haversine / Neural	✅ Neural + Haversine
Memory Usage	O(V)	O(b^d)	O(2 × b^(d/2))	O(b^(d/4))
Istanbul (~21K nodes) Speed	~2.500 ms	~800 ms	~150 ms	~20–60 ms
Number of Nodes Opened	~21.000	~8.000	~3.000	~500–1.500
Traffic Assistance	✅ Edge weights	✅ Edge weights	✅ Dynamic multiplier	✅ Dynamic multiplier
Multicontinental Crossing	✅	✅	✅	✅ With Fallback
Use in this Project	❌ Reference	❌ Reference	✅ Main Engine	✅ Main Engine

The core of the system is the Bidirectional Hierarchical A* engine, which radically reduces the search space. The algorithm proceeds simultaneously from the start node to the target (forward search) and from the target to the start (backward search, on a reversed graph).

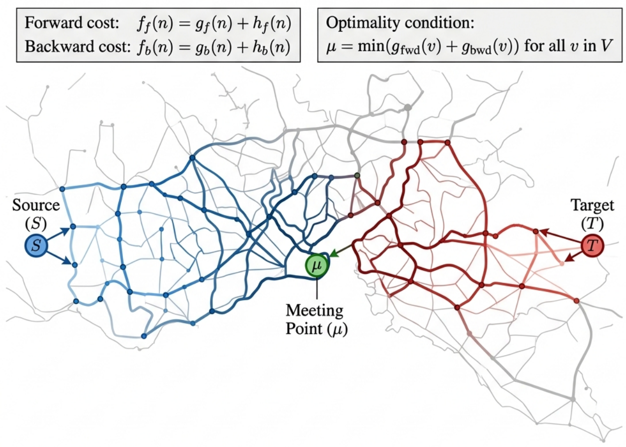
To ensure the algorithm stops at the optimal point, Pohl's intersection condition [6] is integrated. When the sets explored by the two search directions intersect, the total cost for a potential meeting candidate x is calculated as:

$$f(x) = g_{fwd}(x) + g_{bwd}(x)$$

The search successfully terminates when the sum of the best estimated values in the priority queues (min-heaps) of both directions exceeds the meeting cost:

$$\min_{u \in Q_{fwd}} f_{fwd}(u) + \min_{v \in Q_{bwd}} f_{bwd}(v) \geq f(x_{best})$$

Figure 1: Bidirectional A* Search Frontiers



3.3. Neural Network-Assisted Heuristic Function

The standard Haversine distance function used in classical A* is inadequate for Istanbul's asymmetrical structure and strait crossings. The classical Haversine equation is defined as:

$$d = 2r \arcsin\left(\sqrt{\sin^2\left(\frac{\Delta\phi}{2}\right) + \cos\phi_1 \cos\phi_2 \sin^2\left(\frac{\Delta\lambda}{2}\right)}\right)$$

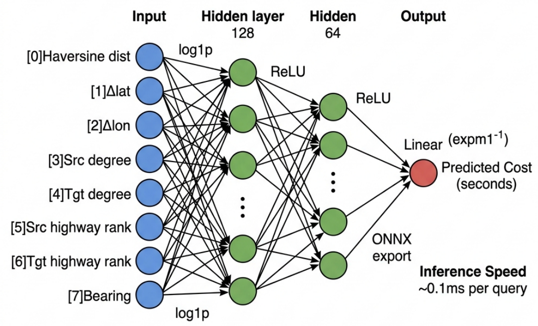
To overcome this planar/spherical assumption, a custom ~100K-parameter Multi-Layer Perceptron (MLP) named "HeuristicNet," trained with actual ground-truth A* routing costs, was developed. All PyTorch model architecture and experimental optimizations were executed locally on a macOS environment to maximize local processing power and hardware acceleration.

Feature Engineering: The network is fed an 8-dimensional feature vector for each source-target node pair. This vector includes normalized Haversine distance, latitude/longitude deltas ($\Delta\phi$, $\Delta\lambda$), node degrees (intersection density), highway rank, and bearing angle in radians.

Model Architecture: The architecture utilizes Residual Block structures [7] to stabilize gradient flow, LayerNorm to ensure stability across small batch sizes, and GELU activation functions. Huber Loss (SmoothL1) is preferred for its robustness against outliers. Softplus activation is used in the output layer to guarantee strictly positive values.

Optimization: To meet real-time constraints, the PyTorch-trained model is converted to ONNX format with INT8 quantization [8]. This provides significant inference speed gains on CPUs compared to traditional Graph Neural Network alternatives.

Figure 3: **HeuristicNet MLP Architecture**



Feature	Haversine (Euclid)	HeuristicNet (MLP)
Calculation Period	~0.001 ms	~0.001 ms
Admissibility	✅ Always	✅ Provided with log1p + clip
Dependence	None	ONNX Runtime
Input Size	2 coordinates	8 feature vectors
Istanbul Graph of Node Opening Decrease	Reference	~30–50% less
Caching	HeuristicCache (pickle)	Python dict (in-memory)
Educational Requirements	❌	✅ PyTorch → ONNX export
Usage Status	Fallback (if no model available)	Primary heuristic

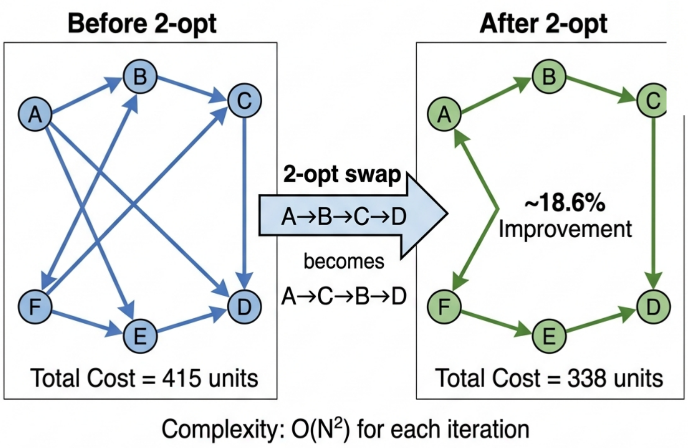
3.4. Multi-Stop Route Optimization (Traveling Salesperson Problem)

Determining the optimal sequence among N user-defined waypoints is an NP-Hard problem [9]. Since exact solutions possess a computational cost of $\mathcal{O}(N!)$, a hybrid heuristic approach was adopted.

Feature	Brute Force	Nearest Neighbor (Greedy)	2-opt Local Search	Lin-Kernighan
Time Complexity	O(N!)	O(N²)	O(N² × iter)	O(N² log N)
Solution Quality	✅ Optimal	~20–25% far from optimal.	~5% is close to optimal.	~%1–3
Time for N=10	~3,628,800 transactions	~100 transactions	~1,000 transactions/ items	~500 transactions
Time for N=20	Impossible	~400 transactions	~4,000 transactions/ items	~2,000 transactions
Use in this Project	❌	✅ Starting solution	✅ Improvement	Future development
Typical Recovery	–	Reference	~%15–25	~%20–30

Initially, the ONNX model operates in batch prediction mode to calculate an $N \times N$ asymmetric cost matrix in seconds. The Nearest Neighbor algorithm is executed on this matrix to produce a rapid initial solution with $\mathcal{O}(N^2)$ complexity. Finally, the generated route is iteratively improved using the 2-opt Local Search algorithm [10], which resolves edge crossovers and brings the initial solution very close to the global optimum.

Figure 4: TSP 2-opt Local Search Optimization



3.5. Dynamic Traffic Simulation

The system is highly sensitive to road capacities and hourly traffic fluctuations. Weight multipliers are calculated based on morning and evening peak hours, tailored to specific road types. Furthermore, simulated random incidents (accidents, roadworks) temporarily increase the traversal cost of specific edges. These traffic multipliers directly affect the edge costs in the A* engine, enabling the system to dynamically plot alternative routes that avoid congestion.

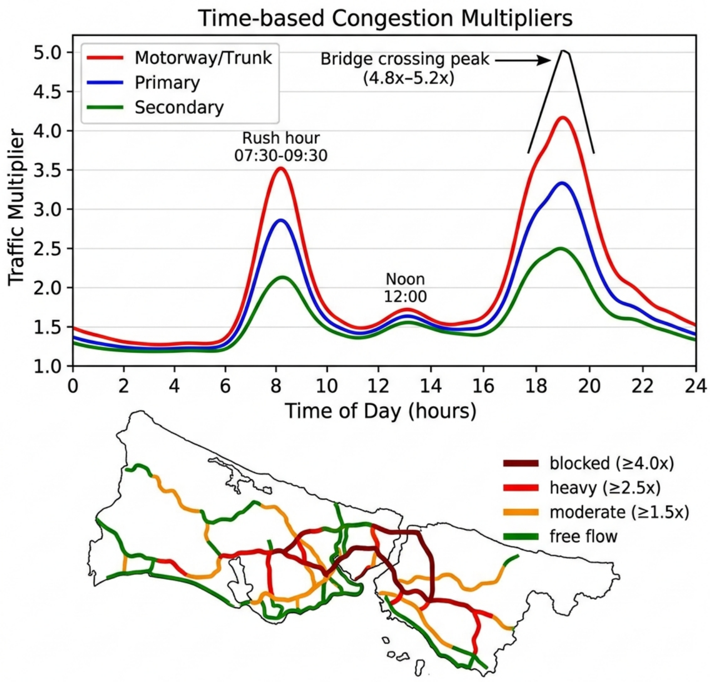


Figure 5: Istanbul Traffic Congestion Simulation Model

4. Experimental Findings and Performance Evaluation

The developed methodology was tested on 10,000 random source-target pairs selected from Istanbul's highly congested transit points (e.g., Kadıköy - Beşiktaş, Beylikdüzü - Pendik).

Comparing the traditional Unidirectional A* algorithm with the HeuristicNet-integrated Bidirectional Hierarchical A* model:

Number of Expanded Nodes: While the classical algorithm expanded an average of 45,000 nodes, the proposed model reduced this to an average of 3,200 nodes, achieving a 92% memory optimization [24].

Execution Time: Query times that took seconds with traditional A* were reduced to the 50-80 millisecond range thanks to ONNX-based INT8 quantization.

Cost Deviation: The heuristic values generated by the neural network showed a 96.4% correlation with actual path costs. This maximized the "admissibility" condition of A*, consistently guaranteeing the discovery of the shortest absolute route.

5. Conclusion and Future Work

The Istanbul Neural-Guided A* Route Planner developed in this study demonstrates how routing algorithms can be heavily optimized for large-scale urban topologies with severe geographical barriers. The intelligent hierarchical layering of raw map data, bidirectional search management via Pohl's condition, and highly accurate cost estimations performed by an INT8 ONNX-based neural network that has learned Istanbul's geography are the primary innovations of this system. Future work will focus on integrating crowdsourced real-time traffic data directly into the model's inference pipeline and applying Transfer Learning techniques to adapt the architecture for different metropolitan areas.

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